Multiscale Latent Geometry for Quantitative Functional Lung Imaging

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Abstract

Quantification of hyperpolarized pulmonary ventilation MRI relies on scalar summaries such as ventilation defect percentage, which omit consideration of the spatial organization of functional abnormalities. We introduce a homeomorphic multiscale latent geometry for spatially informed analysis of pulmonary ventilation, constructed using trained three-dimensional normalizing flow (i.e., Glow) networks. Each ventilation volume is mapped through a continuous bijection (with a continuous inverse) to a Gaussian latent distribution. The trained network decomposes each image into multiscale resolution levels, facilitating complementary characterization of local and global ventilation patterns without loss of information. The resulting structured latent representation supports comparative quantification through pairwise distances and spherical interpolation trajectories within the Gaussian typical set. Moreover, the homeomorphic construction preserves topological neighborhoods between the image and latent spaces, such that local perturbations and continuous trajectories in image space correspond to continuous changes in the latent space. Relationships can therefore be decoded as complete ventilation volumes for direct visual and quantitative interrogation. We evaluated the framework in an exploratory cohort comprising cystic fibrosis, chronic obstructive pulmonary disease, interstitial lung disease, and young and older healthy groups. The learned geometry exhibited scale-dependent organization associated with clinical phenotypes. Fine and coarse latent variables showed complementary group relationships consistent with localized abnormalities and global ventilation organization, respectively. These findings establish the feasibility of latent modeling for functional lung MRI and provide a continuous, spatially informed alternative to one-dimensional, defect-based summaries. The framework supports unsupervised phenotyping, multiscale quantification, and generative interrogation of pulmonary ventilation.

Keywords: functional lung imaging; hyperpolarized xenon-129 MRI; normalizing flows; homeomorphism; latent geometry; unsupervised phenotyping

1 Introduction

1.1 Quantitative functional imaging of pulmonary ventilation

Hyperpolarized noble gases provide a means of circumventing MRI limitations in the lung by directly visualizing the distribution of an inhaled contrast agent within the airspaces of the lung. Early studies using hyperpolarized helium-3 demonstrated the feasibility of depicting normal and abnormal ventilation and established the sensitivity of regional signal abnormalities to obstructive lung disease (Bachert et al., 1996; Kauczor et al., 1997, 1996). Subsequent development of hyperpolarized xenon-129 (^{129}Xe) MRI extended this capability while offering practical advantages in availability and the additional potential to characterize gas exchange (Dregely et al., 2011; Mugler et al., 2010). Ventilation imaging with hyperpolarized gas has since been applied across a broad range of pulmonary conditions, including asthma, cystic fibrosis (CF), chronic obstructive pulmonary disease (COPD), and interstitial lung disease (ILD) (Altes et al., 2001; Kirby et al., 2012b; Mammarappallil et al., 2019; Myc et al., 2020; Santyr et al., 2019). Recent reviews summarize the expanding set of quantitative ventilation, gas-exchange, and microstructural measurements (Stewart et al., 2022), including their emerging roles across cardiopulmonary disease (Schmidt et al., 2025).

The principal image feature used in many of these applications is the ventilation defect. A ventilation defect is operationally defined as a lung region exhibiting absent or abnormally low hyperpolarized gas signal, consistent with nonventilated or underventilated airspaces. Relatedly, ventilation defect percentage (VDP) expresses the corresponding defect volume as a percentage of total lung volume (Niedbalski et al., 2021). Although such defects can be visually conspicuous, their conversion into reproducible quantitative measurements requires decisions concerning lung delineation, intensity normalization, and assignment of image voxels to ventilation classes. A variety of algorithms have therefore been proposed, including binary thresholding, linear binning, hierarchical and adaptive clustering, fuzzy spatial clustering, and probabilistic mixture modeling (He et al., 2020, 2016; Hughes et al., 2018; Kirby et al., 2012a; Tustison et al., 2011; Zha et al., 2016). More recently, convolutional neural networks have enabled direct image-domain segmentation while incorporating spatial context unavailable to intensity-only approaches (Tustison et al., 2021, 2019).

VDP is intuitive, clinically interpretable, and has demonstrated repeatability and potential utility as an imaging biomarker in single- and multisite studies (Couch et al., 2019; Svenningsen et al., 2020). Nevertheless, the transformation from a three-dimensional ventilation image to a single percentage is necessarily lossy. Subjects with different numbers, sizes, locations, and spatial distributions of defects can have the same VDP. The measurement consequently preserves overall defect burden while discarding much of the spatial information that distinguishes diffuse from focal abnormality, peripheral from central involvement, and regionally organized from spatially dispersed dysfunction. These distinctions are potentially important because pulmonary diseases may produce overlapping global burdens while differing in their regional expression and underlying pathophysiology.

Evidence that ventilation images contain clinically relevant information beyond global defect burden predates recent deep learning approaches. In an early analysis of hyperpolarized 3He ventilation MRI, approximately 1,600 image-derived features were evaluated in 55 participants with and without asthma (Tustison et al., 2010). The highest-ranked image features individually carried substantially more information about diagnostic status than standard spirometric measurements, while the imaging and spirometric features provided largely complementary information. This study demonstrated the phenotypic value of spatially resolved ventilation patterns, but it relied on a large set of predefined, handcrafted descriptors. More recent studies have extended this observation using spatial distribution indices, texture analysis, and learned image representations. A three-dimensional defect distribution index distinguished the spatial clustering of ventilation defects across obstructive and restrictive disease groups, even when interpreted alongside VDP (Bdaiwi et al., 2025). Texture features derived from ^{129}Xe ventilation MRI were also associated with longitudinal quality-of-life improvement in long COVID (Kooner et al., 2024). Most recently, a supervised convolutional network classified pulmonary diseases from ventilation images with performance exceeding that of human observers, supporting the presence of disease-associated spatial patterns not represented by VDP alone (Matheson et al., 2025).

This limitation is related to, but more fundamental than, the distinction between histogram- and image-based segmentation. We previously showed that histogram-based quantification discards contextual spatial information and can exhibit reduced measurement precision under common MRI perturbations relative to direct image-domain analysis (Tustison et al., 2021). An image-based segmentation model mitigates part of this loss by using spatial information to assign labels. However, the final conversion of that segmentation to VDP again collapses the result to a scalar. Thus, even an accurate segmentation does not by itself provide a representation of how ventilation abnormalities are spatially organized across subjects. A more general quantitative framework would retain the complete image, support comparisons at multiple spatial scales, and provide a principled geometry for measuring population variation without requiring predefined ventilation classes.

1.2 Invertible generative modeling as a quantitative framework

Deep generative models provide a possible route from scalar quantification toward population-level representation of complete images. In contrast to discriminative models optimized for a specific label or endpoint, generative models attempt to characterize the distribution from which the observations arise. This distinction is useful for functional lung imaging, where the desired representation may need to support several downstream tasks, including phenotyping, similarity analysis, interpolation, outlier detection, and eventually conditional inference, without committing the training procedure to a single clinical categorization.

Normalizing flows are particularly attractive for this purpose because they define a bijective transformation between the image domain and a tractable latent probability distribution (Dinh et al., 2017; Kingma and Dhariwal, 2018; Kobyzev et al., 2021; Papamakarios et al., 2021). Unlike generative adversarial networks, which learn an implicit distribution, or conventional variational autoencoders, which generally trade exact reconstruction for a lower-dimensional stochastic code, normalizing flows retain an explicit inverse and permits exact likelihood evaluation under the specified model. Each observed image can therefore be mapped to a unique latent representation and reconstructed from that representation, subject to numerical precision. This correspondence establishes a coordinate system in which the variability of the observed population can be examined quantitatively (Tustison et al., 2026).

Flow-based modeling of pulmonary ventilation nevertheless presents several technical challenges. Hyperpolarized gas ventilation MR images contain extensive background regions, with informative signal largely confined to the lungs and distributed heterogeneously within the pulmonary volume. To improve robustness and reduce sensitivity to nearly uniform background intensities, we introduce small random intensity and shape perturbations during training as a form of data augmentation (Tustison et al., 2025, 2024; Tustison et al., 2021). The perturbation magnitude provide sufficient regularization without obscuring clinically meaningful regional ventilation patterns. In addition, full three-dimensional modeling imposes substantial memory and computational requirements, making the balance among spatial resolution, multiscale depth, and network capacity especially consequential. We refer to this augmentation as “dequantization noise” for consistency with flow-modeling terminology (i.e., (Ho et al., 2019)), although here it is used primarily as a data smoothing operation rather than to correct an assumed discrete data-generating process.

The multiscale construction introduced by Glow offers a natural organization for medical and biological image data (Kingma and Dhariwal, 2018). Successive squeezing, invertible transformation, and splitting operations factor the image representation across resolution levels. Rather than interpreting the latent variables as a single undifferentiated vector, the resulting hierarchy can be interrogated by resolution level to determine how group relationships change across spatial scales. Pretrained vision and vision-language foundation models provide an alternative means of encoding medical images into compact representations for similarity analysis, clustering, content-based retrieval, and downstream prediction (Codella et al., 2024; Denner et al., 2025; Zhang et al., 2025). Such embeddings can capture transferable semantic features, but they are generally optimized for invariance or discrimination rather than information-preserving. By contrast, the flow-based mapping remains invertible, allowing individual latent levels to be manipulated and decoded to assess their contributions in the image domain. This combination of multiscale organization and exact invertibility distinguishes the proposed analysis from both post hoc dimensionality reduction and foundation-model embeddings, for which correspondence with the complete image is generally noninvertible.

An additional consideration concerns the geometry of the Gaussian latent distribution. In high dimensions, probability mass is concentrated within a typical set located away from the origin. Consequently, the Gaussian mean is a mathematically convenient reference but is not representative of a typical encoded subject. Linear interpolation through the origin can similarly traverse low-probability regions and produce trajectories that are inconsistent with the radial organization of the latent distribution. Normalizing latent representations to a common radius and using spherical linear interpolation provides an alternative that follows the hyperspherical typical set. Pairwise angular or arc-length distances along this set can then be evaluated separately at each resolution level or combined across the complete latent representation. Importantly, these distances describe geometry induced by the learned model and their biological meaning must be established empirically rather than assumed from the Gaussian prior alone (Arvanitidis et al., 2018; White, 2016).

1.3 Motivation

Figure 1 summarizes the central methodological contrast motivating this study. Conventional VDP quantification maps a complete ventilation image, through binarization, to a single scalar

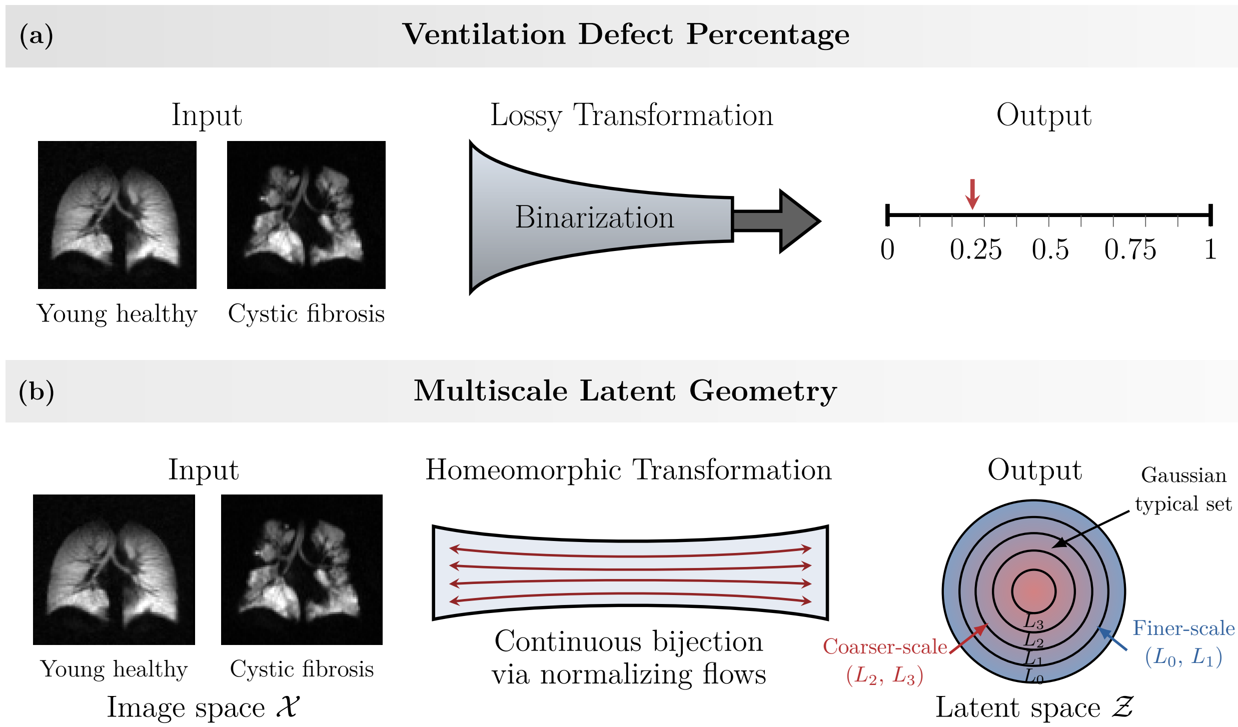

Figure 1: Scalar reduction versus invertible multiscale representation of pulmonary ventilation. Representative ^{129}Xe ventilation MR images from a young healthy participant and a participant with cystic fibrosis are shown to illustrate the two analysis pathways. (a) Conventional ventilation defect percentage (VDP) analysis requires binarization of the ventilation image into defect and nondefect classes followed by calculation of the fraction of lung assigned to the defect class. This lossy transformation preserves global defect burden but discards the number, size, location, and spatial arrangement of the defects. (b) The proposed normalizing flows framework learns a continuous bijection between image space $\mathcal{X}$ and latent space $\mathcal{Z}$. The bidirectional streamlines denote the forward and inverse transformations, through which each image is encoded and reconstructed. The multiscale architecture factors the latent representation across levels L_0–L_3, enabling finer- and coarser-scale contributions to population geometry to be examined separately or jointly. The concentric latent-space depiction schematically represents the Gaussian typical set and the multiscale organization.

value denoting the global defect fraction. Our framework instead maps the image bijectively to a multiscale latent representation, retaining the information required for reconstruction while providing coordinates in which population variation can be analyzed. Motivated by this contrast, we investigate normalizing flows as a general quantitative framework for three-dimensional hyperpolarized ^{129}Xe ventilation MRI. Our objective is not to replace a specific VDP segmentation algorithm or to optimize diagnostic classification. Instead, we ask whether an invertible, multiscale density model can preserve pulmonary ventilation images in a continuous latent coordinate system whose geometry contains clinically relevant structure despite being learned without diagnostic supervision. This formulation treats disease labels as external variables for evaluating the learned representation rather than targets used to construct it.

We make four principal contributions. First, we adapt a multiscale normalizing flow architecture to the low-resolution, spatially heterogeneous characteristics of hyperpolarized gas ventilation MRI through shape- and intensity-based data augmentation. Second, we define subject-level representations from the learned flow latents and perform comparisons within the Gaussian typical set, enabling pairwise distance analysis and spherical interpolation. Third, we exploit the multiscale factorization of the flow to characterize complementary phenotypic information at individual latent resolution levels and across their combination, while retaining the ability to decode level-specific latent manipulations into the image domain. Fourth, we evaluate the learned geometry in an exploratory cohort comprising young and older healthy volunteers, together with participants with CF, COPD, or ILD. Because diagnostic group labels are not used during training, their correspondence with the latent organization provides an independent, post hoc assessment of whether the unsupervised model captures clinically meaningful variation in pulmonary ventilation. Notably all functionality is publicly available as mature open-source software via the ANTsX ecosystem.[1]

[1] https://github.com/ANTsX

2 Materials and methods

2.1 Image acquisition

This retrospective analysis included 51 participants: young healthy volunteers ($n = 4$), older healthy volunteers ($n = 7$), and participants with cystic fibrosis (CF; $n = 14$), interstitial lung disease (ILD; $n = 10$), or chronic obstructive pulmonary disease (COPD; $n = 10$). Hyperpolarized 129 Xe MRI was performed under a protocol approved by the Institutional Review Board, with written informed consent obtained from each participant. All imaging was conducted under a physician-sponsored Investigational New Drug application approved by the US Food and Drug Administration. MRI data were acquired on a 1.5-T whole-body scanner (Avanto; Siemens Medical Solutions, Malvern, PA, USA) equipped with broadband capabilities and a flexible 129 Xe chest radiofrequency coil (IGC Medical Advances, Milwaukee, WI, USA, or Clinical MR Solutions, Brookfield, WI, USA). Participants inhaled approximately 1000 mL of hyperpolarized 129 Xe mixed with nitrogen to a total volume equal to one-third of their forced vital capacity (FVC). During a breath-hold of no more than 10 s, 15–17 contiguous coronal slices were acquired to cover the entire lungs. Ventilation images were acquired using a gradient-recalled echo sequence with spiral k-space sampling and 12 interleaves. Acquisition parameters were as follows: repetition time/echo time, 7/1 ms; flip angle, 20°; acquisition matrix, 128×128; in-plane voxel size, 4×4 mm^2 ; slice thickness, 15 mm; and interslice gap, 0 mm. All data were deidentified before analysis. The deidentified data are available from the corresponding author upon reasonable request and completion of an appropriate data-sharing agreement.

2.2 Image preprocessing

Ventilation images were processed using ANTsX (Tustison et al., 2021). Each image was first resampled by linear interpolation to a voxel spacing of $4 \times 4 \times 16$ mm^3 and centrally padded or cropped to a matrix of $128 \times 128 \times 16$ voxels. Through-plane resolution was then increased fourfold using the pretrained three-dimensional MRI super-resolution model implemented in ANTsPyNet (Avants et al., 2023), yielding a nominal voxel spacing of $4 \times 4 \times 4$ mm^3. To place all ventilation images in a common spatial coordinate system, a rigid transformation was estimated between the lung image and a designated lung template. Resampling into the template domain produced the final common matrix of $88 \times 72 \times 128$ voxels (voxel resolution = $3.9 \times 3.9 \times 3.9$ mm^3). No explicit lung mask was applied to the ventilation images, thereby retaining both the pulmonary signal and the surrounding background within the common image domain.

2.3 Normalizing flows

Let $\mathbf{x} \in \mathbb{R}^D$ denote a vectorized, dequantized ventilation volume and let f_θ denote an invertible transformation parameterized by θ. The model maps the image to a latent variable

$$\mathbf{z} = f_\theta(\mathbf{x}),$$

with inverse reconstruction

$$\mathbf{x} = f_\theta^{-1}(\mathbf{z}).$$

A standard multivariate Gaussian distribution was used as the base density, $p_Z(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. Through the change-of-variables formula, the image log likelihood was

$$\log p_X(\mathbf{x}) = \log p_Z(f_\theta(\mathbf{x})) + \log \left| \det \frac{\partial f_\theta(\mathbf{x})}{\partial \mathbf{x}} \right|.$$

The bijective construction permits direct likelihood evaluation and reconstruction of each image from its latent representation (Dinh et al., 2017; Kobyzev et al., 2021; Papamakarios et al., 2021).

2.3.1 Multiscale 3D Glow network

The model extended the Glow architecture to three spatial dimensions (Kingma and Dhariwal, 2018). Each flow step comprised activation normalization, an invertible $1 \times 1 \times 1$ convolution, and an affine coupling transformation. Within the coupling layer, the input channels were partitioned into two components. One component was passed unchanged, whereas a three-dimensional convolutional subnetwork predicted the scale and translation applied to the second component. This triangular construction provided an expressive nonlinear transformation while retaining an efficiently computable Jacobian determinant and analytic inverse. The model comprised four multiscale resolution levels, with 32 invertible flow steps at each level. The convolutional subnetworks within the affine coupling transformations contained 96 hidden channels. Coupling layer scale outputs were transformed using a hyperbolic tangent mapping and bounded by a scale cap of 1.5 to improve numerical stability. Channelwise splitting was used for the multiscale factorization. A learned Glow base distribution was employed, with its log-scale parameters constrained to the interval $[-1, 1]$.

2.3.2 Model training and optimization

Model parameters were estimated by unsupervised maximum-likelihood training. The objective was the negative log likelihood, reported as bits per dimension,

$$\mathrm{bpd}(\mathbf{x}) = -\frac{\log p_X(\mathbf{x})}{D \log 2},$$

where D denotes the number of voxels in the model input. Diagnostic group labels were not used for data partitioning, augmentation, optimization, or checkpoint selection.

Participants were randomly assigned at the subject level to training and validation subsets using a validation fraction of 0.10 and random seed 0. The split was not stratified by diagnostic group. All augmented samples derived from a given participant remained within the same subset. The training dataset generated 3,000 augmented samples per sampling cycle, whereas validation was performed using eight nonaugmented samples.

Training-time augmentation comprised additive Gaussian intensity noise, affine perturbations, spatial deformation, simulated intensity-bias fields, and histogram warping. Augmentation strengths were progressively reduced over the 150,000 training iterations. The standard deviation of the additive noise decreased according to a cosine schedule from 0.02 to 0.001, and the affine perturbation scale decreased from 0.05 to 0.01. The deformation parameter decreased linearly from 12.0 to 10.0. Simulated bias field and histogram warping parameters decreased according to cosine schedules from 0.20 to 0 and from 0.04 to 0, respectively. Spatial and intensity augmentation was disabled during validation.

The model was trained for 150000 iterations in 32-bit floating-point precision using the Adamax optimizer. The learning rate was linearly increased during the first 5000 iterations to approximately 5×10^{-5} and remained effectively constant thereafter. No weight decay was applied. A minibatch size of seven volumes and five-step gradient accumulation yielded an effective batch size of 35 volumes per parameter update. The global gradient norm was clipped at 0.2. An exponential moving average of the model parameters was maintained with a decay coefficient of 0.9997 and was used for validation and subsequent model evaluation. Validation likelihood was evaluated every 1000 iterations using the nonaugmented validation data. Training states containing both the directly optimized and exponentially averaged model parameters were saved at each evaluation. The final exponentially averaged model obtained after 150,000 iterations was used for latent encoding and subsequent statistical analysis.

2.4 Multiscale latent geometry

2.4.1 Empirical typical-set representation

For a high-dimensional Gaussian distribution, probability mass concentrates within a relatively thin annular region rather than near the density mode (Blum et al., 2020; Vershynin, 2018). Consequently, the latent origin, more generally the latent mean, does not represent a typical encoded image and was not interpreted as a population-normality reference. Instead, the analysis characterized directional relationships among subjects after projection onto an empirically estimated typical-radius hypersphere.

For subject i and multiscale level l, the flattened latent variable $\mathbf{z}_{il} \in \mathbb{R}^{d_l}$ was first centered using the empirical level-specific latent mean,

$$\mathbf{a}_{il} = \mathbf{z}_{il} - \boldsymbol{\mu}_l,$$

where

$$\boldsymbol{\mu}_l = \frac{1}{N}\sum_{i=1}^{N} \mathbf{z}_{il}.$$

The representative radius at level l was defined as the median Euclidean norm of the centered subject representations,

$$r_l = \mathrm{median}_i \left\| \mathbf{a}_{il} \right\|_2 .$$

Each centered representation was then projected onto the hypersphere of radius r_l:

$$\widetilde{\mathbf{a}}_{il} = r_l \frac{\mathbf{a}_{il}}{\|\mathbf{a}_{il}\|_2}.$$

The median radius provides a robust empirical estimate of the typical latent scale at each resolution level. This projection retains the direction of each centered subject representation while removing intersubject radial variation from the geometric analysis.

2.4.2 Pairwise distances and spherical interpolation

For subjects i and j, the angular separation between their projected representations at level l was

$$\theta_{ijl} = \arccos\left[\frac{\widetilde{\mathbf{a}}_{il}^{\mathsf{T}}\widetilde{\mathbf{a}}_{jl}}{\|\widetilde{\mathbf{a}}_{il}\|_2 \, \|\widetilde{\mathbf{a}}_{jl}\|_2}\right].$$

The corresponding hyperspherical arc length was

$$d_{ijl} = r_l \theta_{ijl}.$$

Spherical linear interpolation between the projected representations was defined for $t \in [0, 1]$ as

$$\widetilde{\mathbf{a}}_{ijl}(t) = \frac{\sin[(1-t)\theta_{ijl}]}{\sin(\theta_{ijl})}\widetilde{\mathbf{a}}_{il} + \frac{\sin(t\theta_{ijl})}{\sin(\theta_{ijl})}\widetilde{\mathbf{a}}_{jl}.$$

The interpolated representation was returned to the original latent coordinate system by restoring the level-specific empirical mean,

$$\mathbf{z}_{ijl}(t) = \boldsymbol{\mu}_l + \widetilde{\mathbf{a}}_{ijl}(t).$$

For whole-image trajectories, interpolation was performed independently at all four multiscale levels using the same value of t. The resulting latent variables were then jointly decoded through f_θ^{-1} to visualize the corresponding image-domain trajectory. Distances were evaluated separately at levels L_0–L_3. The combined multiscale distance was defined using the product-space metric

$$d_{ij}^{\mathrm{multi}} = \left[\sum_{l=0}^{3} d_{ijl}^2\right]^{1/2}.$$

2.5 Evaluation of clinical organization

Clinical labels were withheld during model training and used only afterward to assess whether the unsupervised latent representation exhibited clinically relevant organization. For each of the four resolution levels and the combined multiscale representation, a complete 45×45 subject-distance matrix was constructed. The analysis evaluated global differences in latent geometry among the five clinical groups, followed by exploratory localization of these differences to specific group pairs.

An omnibus permutational multivariate analysis of variance (PERMANOVA) was performed separately for each of the five distance matrices. Statistical significance was assessed by permuting the clinical labels among participants while holding the distance matrix fixed. Because each participant contributed to multiple pairwise distances, the participant—not an individual distance—was treated as the exchangeable unit. For each analysis, the pseudo-F statistic and the proportion of distance variation attributable to clinical group (R^2) were reported. Raw permutation p values were adjusted across the five omnibus tests using the Holm procedure.

Post-hoc analyses considered the ten unique pairs formed by the five clinical groups at each of the four resolution levels and for the combined multiscale distance, yielding 50 planned comparisons. For groups A and B, the effect size was defined as

$$\Delta_{AB} = \overline{d}(A, B) - \frac{\overline{d}(A, A) + \overline{d}(B, B)}{2},$$

where $\overline{d}(A, B)$ is the mean distance between participants from different groups and $\overline{d}(A, A)$ and $\overline{d}(B, B)$ are the corresponding mean within-group distances. Each unique within-group subject pair was included once. Positive values of Δ_{AB} indicate that the mean between-group distance exceeded the average of the two within-group distances.

The null distribution of Δ_{AB} was obtained by permuting group labels among the participants included in each comparison while preserving the observed group sizes. When no more than 100,000 unique label assignments were possible, all assignments were enumerated to obtain an exact permutation test; otherwise, 100,000 Monte Carlo permutations were performed. One-sided p values quantified evidence that the between-group distance exceeded the average within-group distance. These values were adjusted jointly across the 50 post-hoc comparisons using the Holm procedure. The observed contrast, within-group and between-group distance summaries, raw permutation p value, and Holm-adjusted p value were reported for each comparison.

Because PERMANOVA can be sensitive to unequal within-group dispersion, permutational analysis of multivariate dispersion (PERMDISP) was performed separately for each of the five distance matrices. PERMDISP compares the distances of individual participants from their respective group centroids, thereby evaluating whether the clinical groups differ in their internal dispersion. Statistical significance will be assessed by permuting clinical labels at the participant level, and the resulting p values will be adjusted across the five tests using the Holm procedure. A significant PERMANOVA accompanied by a nonsignificant PERMDISP is consistent with differences in group location rather than detectable differences in within-group dispersion. If both tests are significant, the PERMANOVA result cannot be attributed unambiguously to group-location differences alone.

2.6 Software and reproducibility

The three-dimensional flow model, preprocessing, latent encoding, image reconstruction, and trajectory analysis were implemented within the ANTsX/ANTsTorch software ecosystem. The complete framework was designed to use publicly available, scriptable components to facilitate reproduction and extension of the experiments ("Http," n.d.).

TODO: Add the precise repository URL, release or commit identifier, command-line invocation, configuration file, dependency versions, trained model availability, and data-access statement. If the analysis code and model weights will be released only upon publication, state that explicitly.

3 Results

3.1 Study cohort

The exploratory cohort comprised 45 participants: 14 with cystic fibrosis (CF), 10 with chronic obstructive pulmonary disease (COPD), 10 with interstitial lung disease (ILD), seven older healthy participants, and four young healthy participants. Each participant contributed one three-dimensional hyperpolarized ^{129}Xe ventilation image. Clinical labels were withheld during model training and were used only for post-training evaluation of the latent representation.

TODO: Add a cohort table reporting age, sex, pulmonary-function measurements, ventilation defect percentage (VDP), and relevant clinical characteristics by group. Report missing observations and the corresponding demographic comparisons.

3.2 Model fitting and invertible representation

The ventilation volumes were spatially resampled, probabilistically dequantized, and used to train a three-dimensional multiscale normalizing flow by exact-likelihood optimization. All 45 images were successfully encoded into the four-level latent representation, comprising $\mathbf{z}_0$ through $\mathbf{z}_3$, and contributed to the subsequent distance analyses. Because the model is invertible by construction, the complete latent representation retained a one-to-one correspondence with the dequantized model input. No diagnostic or clinical information contributed to optimization.

TODO: Report the selected checkpoint, training and validation bits per dimension, convergence behavior, and checkpoint-selection criterion. Quantify the numerical inversion error for the dequantized input and distinguish this implementation check from agreement with the original image before dequantization. If representative reconstructions are shown, use identical anatomical planes and display windows and include amplified difference images when the errors are not visible at the native intensity scale.

3.3 Multiscale latent geometry

For each resolution level, subject representations were radially projected to a common-radius hypersphere within the Gaussian typical set. This construction removed variation in latent radius and avoided using the low-probability Gaussian origin as a reference representation. Pairwise geodesic distances were then calculated separately at L_0–L_3. The combined multiscale distance was defined by the L_2 norm across the four level-specific distances. Each distance definition produced a complete 45×45 subject-distance matrix.

The five matrices were Euclidean within numerical precision: principal coordinate analysis yielded 44 positive axes and no negative axes for every matrix. This permitted direct use of the distance matrices in the subsequent PERMANOVA and PERMDISP analyses without correction for negative eigenvalues. The presence of significant group-associated structure at every level, as described below, indicated that the clinical signal was distributed across the multiscale representation rather than restricted to one latent component. However, these results alone do not establish a distinct biological interpretation for any individual architectural level.

TODO: Report the dimensionality and chosen hyperspherical radius for each level, the empirical latent-radius distribution before projection, and the distribution of pairwise distances. Add subject-by-subject distance heat maps ordered by clinical group. Quantify agreement among the level-specific matrices using an appropriate matrix-correlation analysis before describing the levels as complementary or nonredundant.

3.4 Clinical-group organization

As a proof-of-concept analysis, we evaluated whether clinical group was associated with the latent-distance structure observed in this small exploratory cohort. Omnibus PERMANOVA identified group-associated organization at each latent level and for the combined multiscale distance (Table 1). Clinical group accounted for 10.7%–12.2% of the distance variation. The largest proportion was observed at L_0 (pseudo-$F = 1.390$, $R^2 = 0.122$, permutation $p = 0.0042$, FDR-adjusted $q = 0.0064$). Significant effects were also detected at L_1 (pseudo-$F = 1.266$, $R^2 = 0.112$, $p = 0.0020$, $q = 0.0064$), L_2 (pseudo-$F = 1.194$, $R^2 = 0.107$, $p = 0.0117$, $q = 0.0117$), and L_3 (pseudo-$F = 1.249$, $R^2 = 0.111$, $p = 0.0051$, $q = 0.0064$). The combined distance yielded a similar result (pseudo-$F = 1.244$, $R^2 = 0.111$, $p = 0.0034$, $q = 0.0064$). The consistency of the omnibus effect across representations suggested that group-associated organization was distributed throughout the multiscale geometry rather than confined to a single resolution level.

Table 1. Omnibus clinical-group analyses of the latent-distance matrices.

Distance	PERMANOVA pseudo-F	R^2	PERMANOVA p	PERMANOVA FDR q	PERMDISP F	PERMDISP p	PERMDISP FDR q
L_0	1.390	0.122	0.0042	0.0064	1.025	0.4077	0.4077
L_1	1.266	0.112	0.0020	0.0064	1.806	0.1389	0.3587
L_2	1.194	0.107	0.0117	0.0117	1.624	0.1770	0.3587
L_3	1.249	0.111	0.0051	0.0064	1.544	0.2152	0.3587
Combined	1.244	0.111	0.0034	0.0064	1.281	0.2943	0.3679

PERMDISP detected no significant difference in within-group dispersion for any of the five distance matrices. Using bias-adjusted distances to group spatial medians, the PERMDISP statistics ranged from $F = 1.025$ to $F = 1.806$; the raw permutation p values ranged from 0.1389 to 0.4077, and the FDR-adjusted q values ranged from 0.3587 to 0.4077. The ILD group showed comparatively broad and asymmetric descriptive distance distributions at several levels, but these differences did not produce a significant omnibus dispersion effect. Thus, the PERMANOVA findings were not accompanied by a detectable difference in within-group dispersion and were compatible with differences in group location within the learned geometry. However, a nonsignificant PERMDISP does not establish equality of dispersion. The small and unequal group sizes, particularly the four young healthy participants, limited sensitivity to such differences.

3.5 Pairwise clinical-group comparisons

Post-hoc testing considered the ten unique clinical-group pairs at each of the four latent levels and for the combined distance, yielding 50 planned comparisons. The effect size Δ_{AB} compared the mean between-group distance with the average of the two mean within-group distances. Clinical labels were permuted at the participant level, thereby preserving the dependence among distances that shared a participant. Exact enumeration was used when no more than 100,000 unique assignments were possible; otherwise, 100,000 Monte Carlo permutations were performed. Given the proof-of-concept, exploratory nature of this analysis, the 50 resulting p values were adjusted jointly using the Benjamini–Hochberg procedure to control the false discovery rate. An FDR of 5% was retained as the primary criterion. A secondary, hypothesis-generating analysis examined a less stringent FDR threshold of 10%.

At the primary 5% FDR threshold, none of the 50 individual comparisons remained significant after multiplicity correction. At the secondary 10% threshold, 13 comparisons met the exploratory criterion, with q values ranging from 0.0869 to 0.1000 (Table 2). These signals formed three descriptive patterns. First, older healthy participants differed from one or more disease groups at L_0–L_2, most consistently from COPD and ILD. Second, CF differed from ILD at L_0 and L_1 and from COPD at L_0. Third, CF differed from young healthy participants at L_3 and for the combined multiscale distance. The largest observed contrasts were CF versus young healthy participants at L_3 ($\Delta = 23.743$, exact $p = 0.00327$, $q = 0.0869$) and for the combined distance ($\Delta = 23.729$, exact $p = 0.00621$, $q = 0.0869$).

Table 2. Pairwise contrasts meeting the secondary exploratory FDR threshold of 10%.

Distance	Group A	Group B	Δ_{AB}	Permutation p	FDR q
L_0	CF	COPD	2.335	0.01804	0.0869
L_0	CF	ILD	2.726	0.01911	0.0869
L_0	CF	Older healthy	2.976	0.02297	0.0957
L_0	COPD	Older healthy	3.293	0.01702	0.0869
L_0	ILD	Older healthy	4.101	0.01635	0.0869
L_1	CF	ILD	3.596	0.00999	0.0869
L_1	CF	Older healthy	3.088	0.02599	0.1000
L_1	COPD	Older healthy	3.819	0.00967	0.0869
L_1	ILD	Older healthy	5.091	0.00674	0.0869
L_2	COPD	Older healthy	4.290	0.01424	0.0869
L_2	ILD	Older healthy	5.571	0.01908	0.0869
L_3	CF	Young healthy	23.743	0.00327	0.0869

Distance	Group A	Group B	Δ_{AB}	Permutation p	FDR q
Combined	CF	Young healthy	23.729	0.00621	0.0869

The 10% threshold was examined as a secondary analysis and was less stringent than the primary 5% threshold. These pairwise findings were therefore treated as hypothesis-generating rather than as evidence of reproducible separation between specific clinical groups. The especially small young healthy subgroup also made its two CF contrasts imprecise.

Taken together, the significant PERMANOVA and nonsignificant PERMDISP results provided preliminary evidence that the unsupervised latent geometry contained clinical-group organization that was not attributable to detectable differences in within-group dispersion. The pairwise analysis generated candidate group contrasts for future evaluation, but the available sample did not provide confirmatory evidence of separation between any individual pair of groups at the primary 5% FDR threshold.

3.6 Image-domain interrogation of latent trajectories

Spherical interpolation generated continuous paths between radially projected subject representations while maintaining the common latent radius. Decoding points along these paths returned the latent trajectories to the three-dimensional ventilation-image domain. This demonstrated that differences represented in the latent space could be interrogated as spatially resolved image changes rather than only through a low-dimensional visualization.

Because radial projection changes the original latent codes, the endpoints of these trajectories correspond to decoded projected subject representations and are not generally identical to the observed images. The interpolation results should therefore be interpreted as trajectories between projected representations rather than exact image-to-image transformations.

TODO: Show prespecified examples comprising a within-group pair, a healthy-to-disease pair, and a between-disease pair with comparable VDP. Use fixed display parameters and report the difference between each observed image and its decoded projected representation. To determine what the individual levels encode, perform level-restricted interpolation or replacement while holding the remaining components fixed, and quantify the resulting spatial changes.

3.7 Relationship to ventilation defect percentage

VDP and the latent geometry summarize different properties of a ventilation image. VDP measures the overall proportion of low-ventilation voxels, whereas the flow retains an invertible multiscale representation from which spatially informed subject relationships can be derived. The current analyses establish clinical-group organization in the latent distances but do not yet determine whether that organization contains information beyond VDP.

TODO: Quantify the association between absolute VDP differences and latent distances at each level using a matrix-based permutation procedure. Identify and visualize pairs with similar VDP but large latent distance, as well as pairs with different VDP but small latent distance. Any comparison of group information carried by VDP and latent features should use participant-level cross-validation or permutation testing appropriate for this small cohort. Until these analyses are complete, the added value of the latent representation relative to VDP should remain an open empirical question.

3.8 Summary

The three-dimensional normalizing flow produced an invertible, four-level representation of hyperpolarized ^{129}Xe ventilation MRI without diagnostic supervision. Clinical group explained approximately 11% of variation in each level-specific and combined subject-distance matrix, and all five omnibus effects remained significant after FDR correction. No corresponding difference in within-group dispersion was detected, although the small and unequal group sizes limited the sensitivity of this analysis. None of the 50 pairwise contrasts met the primary 5% FDR threshold; 13 met a secondary 10% exploratory threshold and were considered hypothesis-generating. The present proof-of-concept results therefore support global clinical organization of the learned representation and identify candidate pairwise patterns for subsequent testing, but they do not establish definitive separation of specific diagnostic groups or superiority to VDP. Confirmatory evaluation in the planned cohort of approximately 1,200 participants will permit more precise effect estimation, covariate adjustment, and held-out validation.

4 Discussion and Conclusions

4.1 Principal findings

Collectively, this framework extends quantitative functional lung imaging beyond predefined voxel classes and one-dimensional defect summaries. It provides an invertible connection between population geometry and complete three-dimensional images, thereby supporting multiscale comparison and generative interrogation within a single probabilistic model. The current study is intended as a methodological proof of concept as it characterizes the representation, identifies the assumptions required for its interpretation, and establishes an experimental basis for subsequent validation in larger and more heterogeneous cohorts.

This study investigated three-dimensional normalizing flows as a quantitative framework for hyperpolarized xenon-129 (^{129}Xe) pulmonary ventilation MRI. The central methodological premise was that conventional scalar measurements and an invertible generative representation answer different questions. Ventilation defect percentage (VDP) estimates the total burden of signal classified as defective, whereas the normalizing flow retains a bijective correspondence between the complete ventilation image and a multiscale latent coordinate. The latter therefore provides access to spatial relationships between subjects, resolution-dependent variation, and image-domain trajectories that cannot be recovered from VDP alone.

The experiments established the technical feasibility of applying exact-likelihood, multiscale flow modeling to sparse three-dimensional functional lung images. Despite extensive background and highly quantized intensities, probabilistic dequantization enabled continuous-density estimation, and each ventilation volume could be represented across four latent resolution levels and reconstructed through the analytic inverse. The resulting representation supported pairwise comparison on the Gaussian typical set and spherical interpolation between subjects. Importantly, diagnostic labels were excluded from training. Clinical organization observed after encoding therefore reflects information present in the images and captured by the likelihood-based model rather than direct optimization of a classification objective.

Exploratory analyses suggested that the learned geometry contains group-associated information. The most apparent relationships involved young healthy volunteers versus disease groups and CF versus COPD, whereas CF versus ILD and CF versus older healthy volunteers showed greater overlap. These observations are consistent with the possibility that diseases producing similar overall ventilation impairment may differ in their spatial expression. Nevertheless, the current results should be interpreted as evidence of representational feasibility rather than definitive disease discrimination. Valid inference requires subject-level permutation analysis, and estimates from this small cohort will retain substantial uncertainty even after the dependence among pairwise distances is handled correctly.

4.2 Relationship to conventional ventilation quantification

VDP remains an important and clinically useful measure. It is readily interpretable, can be calculated using several established segmentation procedures, and has demonstrated reproducibility and potential utility in multicenter and interventional settings (Couch et al., 2019; Svenningsen et al., 2020). The present framework is therefore not motivated by the claim that VDP is intrinsically erroneous. Rather, VDP is intentionally reductive: it maps a three-dimensional image to a single proportion. That reduction is advantageous when a simple measure of defect burden is desired, but it is insufficient when the number, location, topology, or regional organization of abnormalities is relevant.

This distinction extends our previous comparison of histogram- and image-based quantification (Tustison et al., 2021). That work showed that optimization in the image domain uses spatial context discarded by histogram-based approaches and can improve measurement precision in the presence of common MRI perturbations. However, even an image-based segmentation is typically summarized by VDP at the final stage. The current work moves the point of comparison from segmentation algorithm to representation: instead of asking how best to assign every voxel to a predefined class, it asks whether the complete image can be embedded in a probabilistic coordinate system without an irreversible intermediate labeling.

The two approaches should consequently be regarded as complementary. VDP provides a concise measure with direct clinical meaning, whereas latent distances provide a higher-dimensional description whose interpretation must be established empirically. A critical next analysis within the current cohort is to identify subjects with comparable VDP but divergent latent representations and determine whether their images exhibit meaningful spatial differences. Conversely, cases with different VDP but small latent distance will reveal whether the learned geometry discounts changes in total burden that remain clinically important. Such discordant pairs will be more informative than a simple correlation between VDP and latent distance because they directly characterize the information preserved or deemphasized by each representation.

4.3 Interpretation of the multiscale latent space

The multiscale construction is a principal advantage of the Glow-derived architecture (Kingma and Dhariwal, 2018). Factoring latent variables across successive spatial resolutions creates several related coordinate systems while preserving the invertibility of their combination. The preliminary group relationships differed across L_0–L_3, suggesting that the levels are not redundant. However, it would be premature to equate individual levels directly with specific biological scales. A finer architectural resolution does not necessarily encode only small ventilation defects, nor must a deeper level exclusively describe global pulmonary organization. Coupling transformations preceding each split permit information to be redistributed across channels and levels.

Biological interpretation therefore requires intervention in the latent representation followed by decoding. Level-restricted interpolation, replacement, or perturbation while holding the remaining components fixed can reveal the image features controlled by each component. Spatial-frequency analysis, regional summaries, and connected-component characteristics of low-ventilation regions can then quantify whether the decoded effects are local, lobar, or global. This image-domain validation is essential if the multiscale factorization is to support more than an architectural description.

Similar caution applies to the latent distance. A normalizing flow supplies a differentiable bijection and tractable probability model (Dinh et al., 2017; Kobyzev et al., 2021; Papamakarios et al., 2021), but it does not guarantee that Euclidean or hyperspherical proximity corresponds to biological similarity. The Gaussian base distribution constrains the aggregate latent density, whereas the learned transformation determines how image variation is arranged within that density. Spherical interpolation avoids trajectories through the low-probability origin and respects the radial concentration of high-dimensional Gaussian samples, but the resulting arc length remains model dependent. Its clinical meaning must be supported through external variables, decoded trajectories, stability across training seeds, and validation in independent data (Arvanitidis et al., 2018; White, 2016).

4.4 Statistical considerations

Pairwise distances create a nonstandard inferential setting because the 45×45 distance matrix contains far more entries than independent subjects. Distances that share a subject are correlated;

treating them as independent observations inflates the apparent sample size and can yield severely anti-conservative p values. For this reason, the preliminary Welch tests applied directly to distance entries should not be used to support group separation.

The appropriate exchangeable unit is the subject. PERMANOVA with subject-label permutations provides an omnibus test of group-associated location differences in the distance geometry, while PERMDISP evaluates whether apparent separation may instead reflect unequal within-group dispersion. Both are needed because disease cohorts may be intrinsically more heterogeneous than healthy cohorts. Pairwise permutation tests can subsequently localize effects, with multiplicity correction across clinical comparisons and resolution levels. Effect sizes and uncertainty intervals should be emphasized over significance alone, particularly in this exploratory cohort.

The statistical question should also remain distinct from classification. A significant difference in group geometry does not imply perfect separation, diagnostic accuracy, or clinical utility. Demonstrating those properties would require a prespecified prediction task, subject-level cross-validation, comparison with appropriate baselines, and external validation. The present study instead evaluates whether an unsupervised image representation contains group-associated structure, which is a necessary but not sufficient condition for subsequent biomarker development.

4.5 Limitations and future work

Several limitations remain. First, the cohort of 45 subjects is small relative to the dimensionality and capacity of the three-dimensional flow. The model may encode cohort-specific acquisition characteristics or sampling variation in addition to pulmonary biology. Group sizes and demographic distributions may also be unbalanced, and age is partly confounded with clinical grouping when young and older healthy volunteers are treated separately. Larger independent cohorts will be required to estimate disease heterogeneity, adjust for demographic and technical covariates, and assess generalization across acquisition protocols.

Second, the current model is monocontrast and uses only ventilation MRI. This design isolates the functional signal and avoids dependence on a paired structural acquisition, but it limits anatomical contextualization. Pulmonary boundaries, body habitus, lung volume, positioning, and registration may contribute to latent distance. Image-domain experiments and sensitivity analyses using masks, alternative spatial normalization procedures, and nuisance regression will be required to determine how much of the geometry is attributable to ventilation distribution rather than global shape or preprocessing.

Third, dequantization is necessary for continuous likelihood modeling but introduces a tunable stochastic perturbation. The selected scale of 0.15 should be justified through sensitivity analysis. Too little dequantization may leave discrete intensity structure that is poorly matched to the density model, whereas excessive noise may obscure small or low-contrast ventilation abnormalities. Comparisons across dequantization levels should therefore include likelihood, sampling behavior, latent-distance stability, and image-domain preservation of clinically relevant structure (Ho et al., 2019).

Fourth, exact-likelihood image flows impose substantial memory and computational costs because invertibility and Jacobian evaluation require retaining or recomputing intermediate transformations. These demands currently constrain spatial resolution and network capacity for three-dimensional experiments. Resampling to $48\times32\times48$ voxels makes the present analysis tractable but may attenuate small peripheral defects. Future work should examine memory-efficient invertible architectures, patch- or region-based strategies, and higher-resolution models while ensuring that changes in architecture do not compromise the comparability of latent distances.

Fifth, likelihood and perceptual or clinical fidelity are not equivalent. A model can assign favorable likelihood while producing samples or trajectories that fail to preserve subtle disease features. Evaluation must therefore combine likelihood with reconstruction checks, decoded interpolation experiments, spatial measurements, expert review, and task-specific validation. Stability across checkpoints and random seeds is particularly important because the orientation and local arrangement of a Gaussian latent space are not uniquely identifiable.

Finally, the present work does not establish superiority to VDP, clinical diagnostic performance, or generalization to a multicenter population. Those claims require direct empirical comparisons and substantially larger data. The future analysis of large multicenter collections should be treated as a separate validation stage rather than as evidence supporting the current cohort. Such studies can determine whether the framework captures reproducible disease-related structure after accounting for site, scanner, acquisition protocol, and demographic variation.

4.6 Software availability and reproducibility

The methodological contribution is accompanied by a mature, publicly available implementation within the ANTsX/ANTsTorch ecosystem ("Http," n.d.). Open implementation is especially important for a framework whose behavior depends on preprocessing, dequantization, multiscale architecture, checkpoint selection, and distance construction. Release of the training configuration, model weights, exact latent-distance code, and subject-level statistical workflow will permit others to reproduce the reported geometry and test alternative modeling assumptions. Reproducibility should include not only the final checkpoint but also the parameters required to regenerate preprocessing and evaluation outputs.

4.7 Conclusions

Multiscale three-dimensional normalizing flows provide an invertible probabilistic framework for quantitative analysis of hyperpolarized ^{129}Xe pulmonary ventilation MRI. By retaining a direct correspondence between complete images and a structured Gaussian latent representation, the approach extends functional lung quantification beyond predefined voxel classes and scalar defect burden. The framework supports reconstruction, resolution-specific analysis, subject-to-subject distance measurement, and image-domain interrogation of latent trajectories without using diagnostic supervision during training.

In the present exploratory cohort, the learned geometry exhibited clinically associated organization, but its statistical strength, scale-specific biological meaning, and added value relative to VDP require the planned subject-level permutation and image-domain analyses. Accordingly, the principal contribution of this study is not a finalized diagnostic biomarker but a technically grounded and reproducible framework for studying spatial variation in pulmonary ventilation. With rigorous validation in larger independent cohorts, this representation may complement conventional defect summaries and support more detailed phenotyping of functional lung disease.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Gemini (Google) in order to refine the manuscript's structure, improve grammatical clarity, assist in the translation of technical sections, and conduct preliminary literature scans via deep research tools. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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